

ENTRY 040 1,376 Circuits: the Plateau Was Premature — August 16, 2026 — Status: COMPLETE

Entry 040 — A Fifth and Sixth Growth Batch Reopen the Question Entry 039 Thought Was Closed

790 to 1,376 Circuits, and the ML Model Improved Again -- More Data Was Still the Right Lever

1. Method

Two more independent batches: entry040_grow_short2.py (seed=777, short/mid bins only, 480 new Bell circuits, 240 per chip) and entry040b_grow_far.py (seed=888, far bins + GHZ, 106 more circuits) -- 586 new circuits total, taking the combined dataset from 790 to 1,376. One checkpoint file was corrupted mid-write when a run was interrupted (a truncated JSON string from a kill mid-save); it was recovered by trimming back to the last complete record boundary, losing only the single in-flight circuit rather than the run.

2. The Floor-Collapse Ratio Keeps Narrowing

	790 circuits (Entry 039)	1,376 circuits (Entry 040)
floor-collapse cases	47	70
Kyiv short-route rate	6.0%	5.7%
Sherbrooke short-route rate	11.6%	9.3%
Sherbrooke / Kyiv ratio	1.93x	1.63x

The ratio has now moved 4.7x → 2.05x → 1.93x → 1.63x across four growth rounds -- still trending down, not yet flat. Sherbrooke still collapses more often than Kyiv, but each larger sample makes the gap look smaller than the last one suggested. This hasn't converged as cleanly as Entry 039 read it.

3. Correcting Entry 039: the ML Plateau Was Premature

	790 circuits (Entry 039)	1,376 circuits (Entry 040)
Gradient boosting CV R ²	0.725	0.785
Gradient boosting CV MAE (all)	4.52 pts	3.92 pts
v4.1 floor-collapse MAE	17.59 pts	17.01 pts
Gradient boosting floor-collapse MAE	14.07 pts	13.78 pts

Entry 039 read two similar-looking growth rounds as a plateau and concluded more volume wouldn't help further. This round contradicts that: CV R² rose meaningfully (0.725 to 0.785) and overall MAE improved by over half a point -- real gains, not noise. The lesson isn't that Entry 039 reasoned badly, it's that two data points looking flat isn't enough to call a plateau, especially this early. That's a useful correction to carry into how future entries interpret small numbers of growth rounds.

Entry 040 — Conclusion

More data is still the highest-leverage thing this project can do before touching model architecture. Both headline numbers moved in the same direction as every prior round -- ratio narrowing, ML advantage strengthening -- and the previous entry's read that they'd stabilized turned out to be one round too early. 1,376 circuits is real progress toward the low thousands a GNN would need, and the trend curve, not just the current values, is now informative: nothing here says growth has stopped paying off yet.

4. Next Step

Continue growing the dataset using the same resumable, checkpointed approach (Entries 034, 035, 038, 039, 040 all use the same methodology and can run in further independent batches) until gains genuinely flatten across two OR MORE consecutive rounds -- not one -- before committing to that as the point to stop and move to a GNN architecture.

Research Log — Index Addendum (Entry 040)

Entry	Title	Date	Status
Entry 040	Fifth/Sixth Growth Batches (1,376 circuits); Plateau Retracted	Aug 16, 2026	Complete

Scripts: entry040_grow_short2.py, entry040b_grow_far.py. Data: quantumbridge_data/entry040_short_dataset.json, quantumbridge_data/entry040_far_dataset.json, quantumbridge_data/entry040_combined_dataset.json (1,376 total), quantumbridge_data/entry040_growth_summary.json. Retracts Entry 039's plateau conclusion -- growth is still improving both the floor-collapse ratio estimate and the ML baseline.